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INTRODUCTION

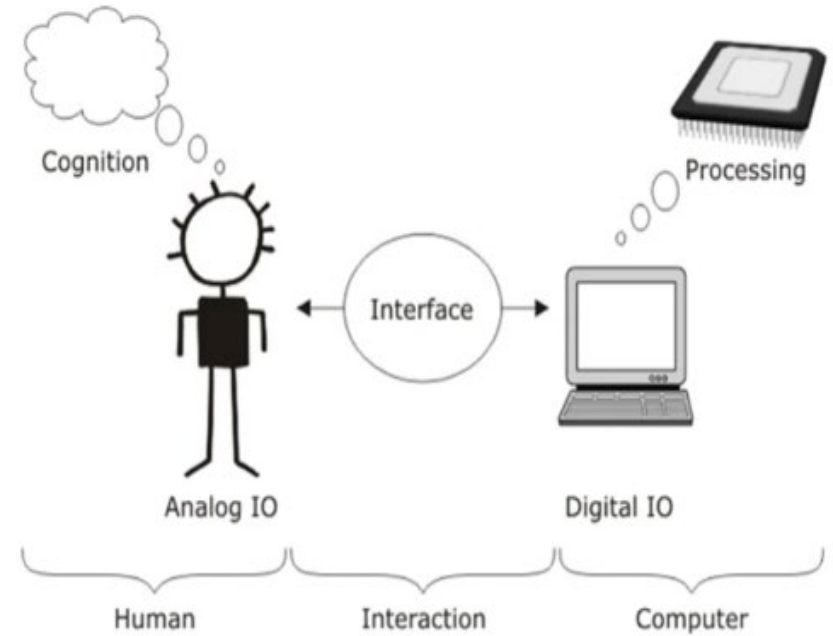


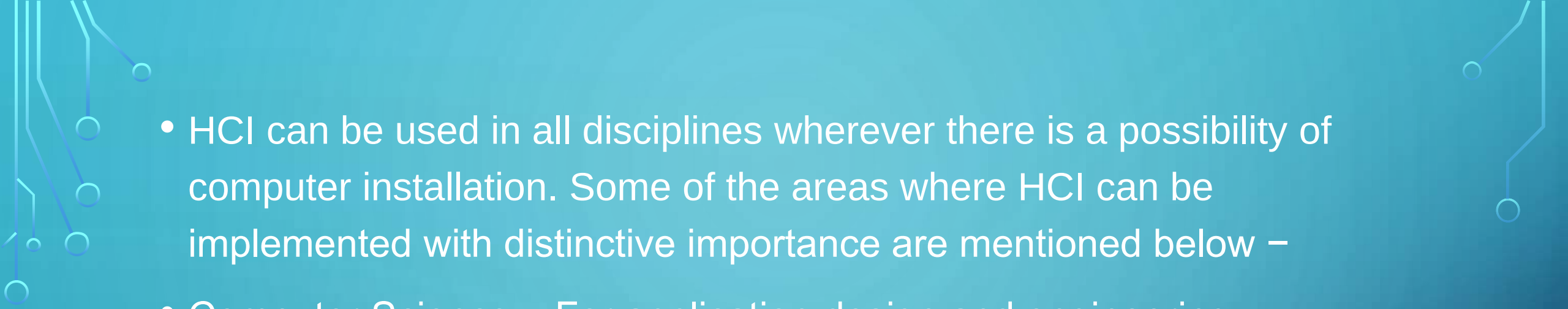
WHAT IS HCI ?

- Human-computer interaction (HCI) is the field of study that focuses on optimizing how users and computers interact by designing interactive computer interfaces that satisfy users' needs. It is a multidisciplinary subject covering computer science, behavioral sciences, cognitive science, ergonomics, psychology, and design principles.
- Human Computer Interface (HCI) is also known as the man-machine studies or man-machine interaction. It deals with the design, execution and assessment of computer systems and related phenomenon that are for human use.

DEFINITION OF HUMAN COMPUTER INTERACTION

- Interaction between user and computer OR relationship between users and computer systems



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- HCI can be used in all disciplines wherever there is a possibility of computer installation. Some of the areas where HCI can be implemented with distinctive importance are mentioned below –
 - Computer Science – For application design and engineering.
 - Psychology – For application of theories and analytical purpose.
 - Sociology – For interaction between technology and organization.
 - Industrial Design – For interactive products like mobile phones, microwave oven, etc.
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THREE (3) FUNDAMENTAL COMPONENTS OF HCI

1. Human User

- People are diverse, unpredictable messy, ill-understood. They have different abilities.

2. Computer system

- Devices are much more complicated and multi-functional then before

3. Interactive process

- Dialogue between human and computer. The interface between humans and the computer they use is crucial to facilitating this interaction



INTERFACE

- Interface is the means by which users interact with the system.

INTERACTION

- Human-computer interaction is two-way communication between a man and a machine

USER INTERFACE

The user interface is the point at which human users interact with a computer, website or application



Defining the User Interface

- The user interface has essentially two components: **input** and **output**.
- **Input** is how a person communicates his or her needs or desires to the computer.
- Example Some common input components
 - keyboard, mouse, trackball, one's finger, and one's voice.
- **Output** is how the computer conveys the results of its computations and requirements to the user.
- Example Some common output components
 - Today, the most common computer output mechanism is the **display screen**, followed by mechanisms that take advantage of a person's auditory capabilities: **voice and sound**.

INPUT / OUTPUT DEVICES FOR INTERACTIVE USERS

- Input devices - send information to a computer system for processing
- Output devices – reproduces or displays the results of that processing
- Most devices are only input devices or output devices as they can only accept data input from a user or output data generated by a computer.

Inputs

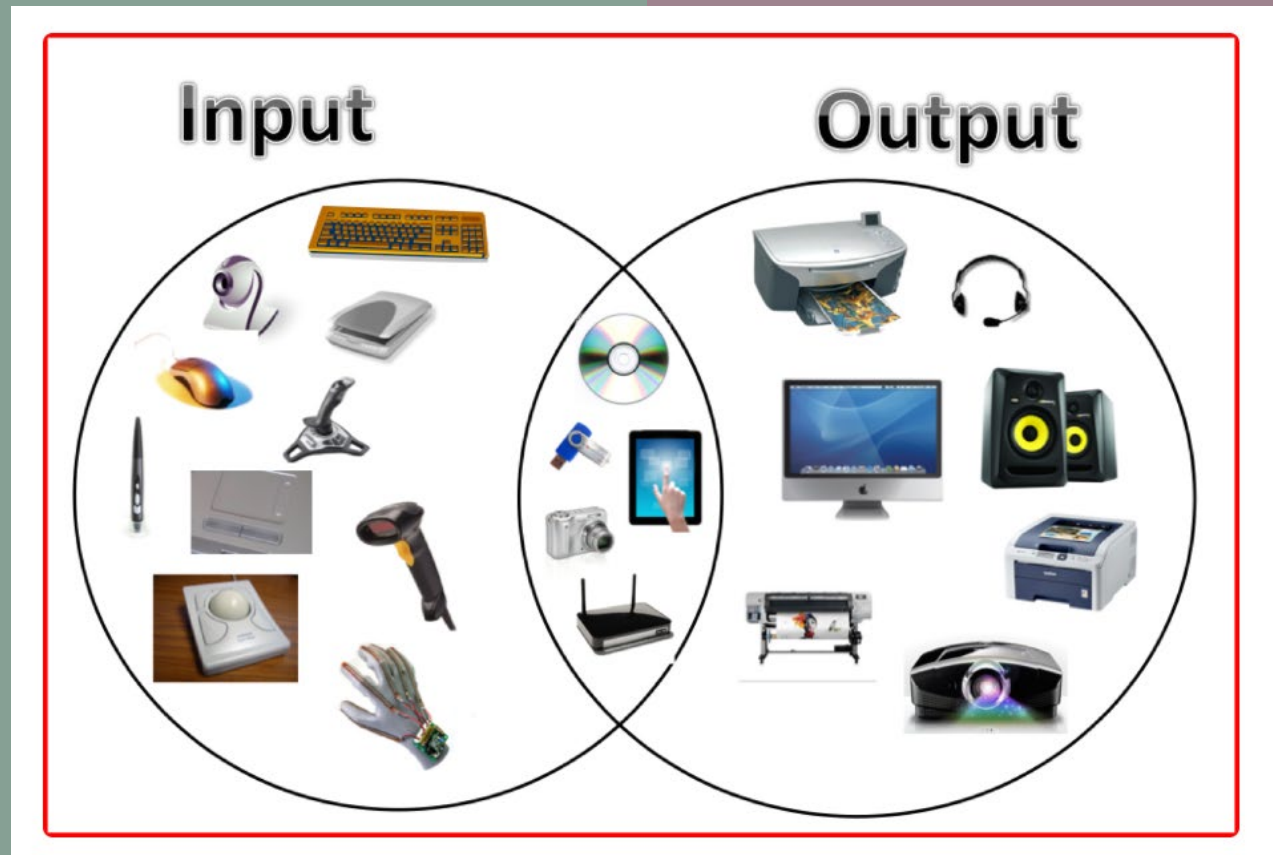
- Keyboard
- Microphone
- Mouse
- Motion sensor
- Heat sensor
- Digital camera
- Touch / Graphics Pad
- Bar code reader

Outputs

- Display
- Speakers
- Printer
- Projector

INPUT / OUTPUT DEVICES FOR INTERACTIVE USERS

- Some devices can accept input and display output and they are referred to as I/O devices.



TYPES OF USER INTERFACES

- GRAPHICAL USER INTERFACE
- MENU-DRIVEN INTERFACE
- COMMAND LINE INTERFACE
- CONVERSATIONAL UI

GRAPHICAL USER INTERFACE

A graphical user interface (GUI) is a digital interface in which a user interacts with graphical components such as icon, buttons and menus



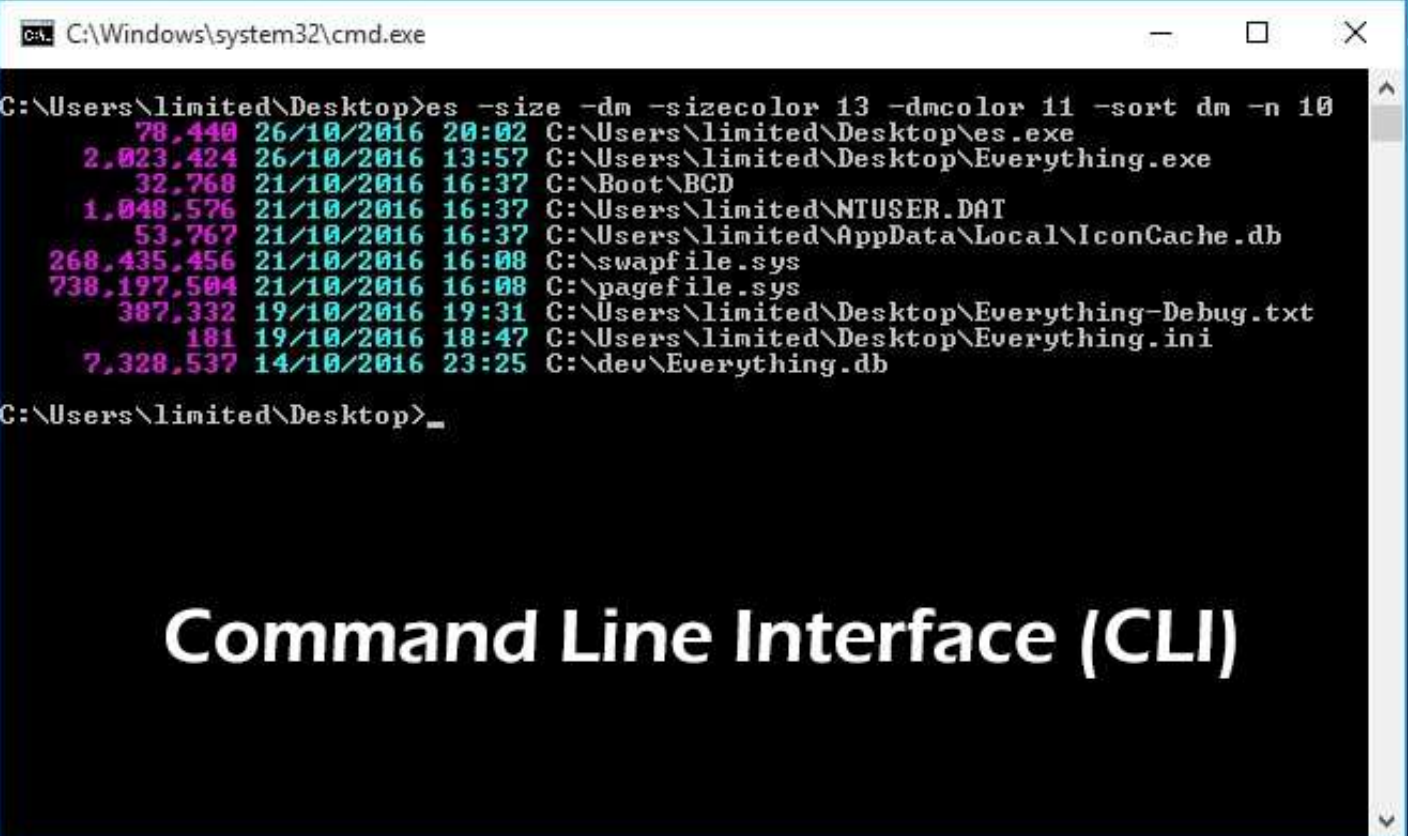
MENU-DRIVEN INTERFACE

The menu-driven user interface provides you with a range of commands or options in the form of a list or menu displayed in full-screen, pop-up, pull-down or drop-down



COMMAND LINE INTERFACE

The Command Line Interface (CLI) is an editing environment that is text-based. It uses specified text (known as commands) to interact with the computer.

A screenshot of a Windows Command Prompt window titled "C:\Windows\system32\cmd.exe". The window shows the output of the command "es -size -dm -sizecolor 13 -dmcolor 11 -sort dm -n 10". The output is a list of files with their size, date, time, and path, color-coded by size. The files listed are: C:\Users\limited\Desktop\es.exe, C:\Users\limited\Desktop\Everything.exe, C:\Boot\BCD, C:\Users\limited\NTUSER.DAT, C:\Users\limited\AppData\Local\IconCache.db, C:\swapfile.sys, C:\pagefile.sys, C:\Users\limited\Desktop\Everything-Debug.txt, C:\Users\limited\Desktop\Everything.ini, and C:\dev\Everything.db. The prompt "C:\Users\limited\Desktop>_" is visible at the bottom.

```
C:\Windows\system32\cmd.exe

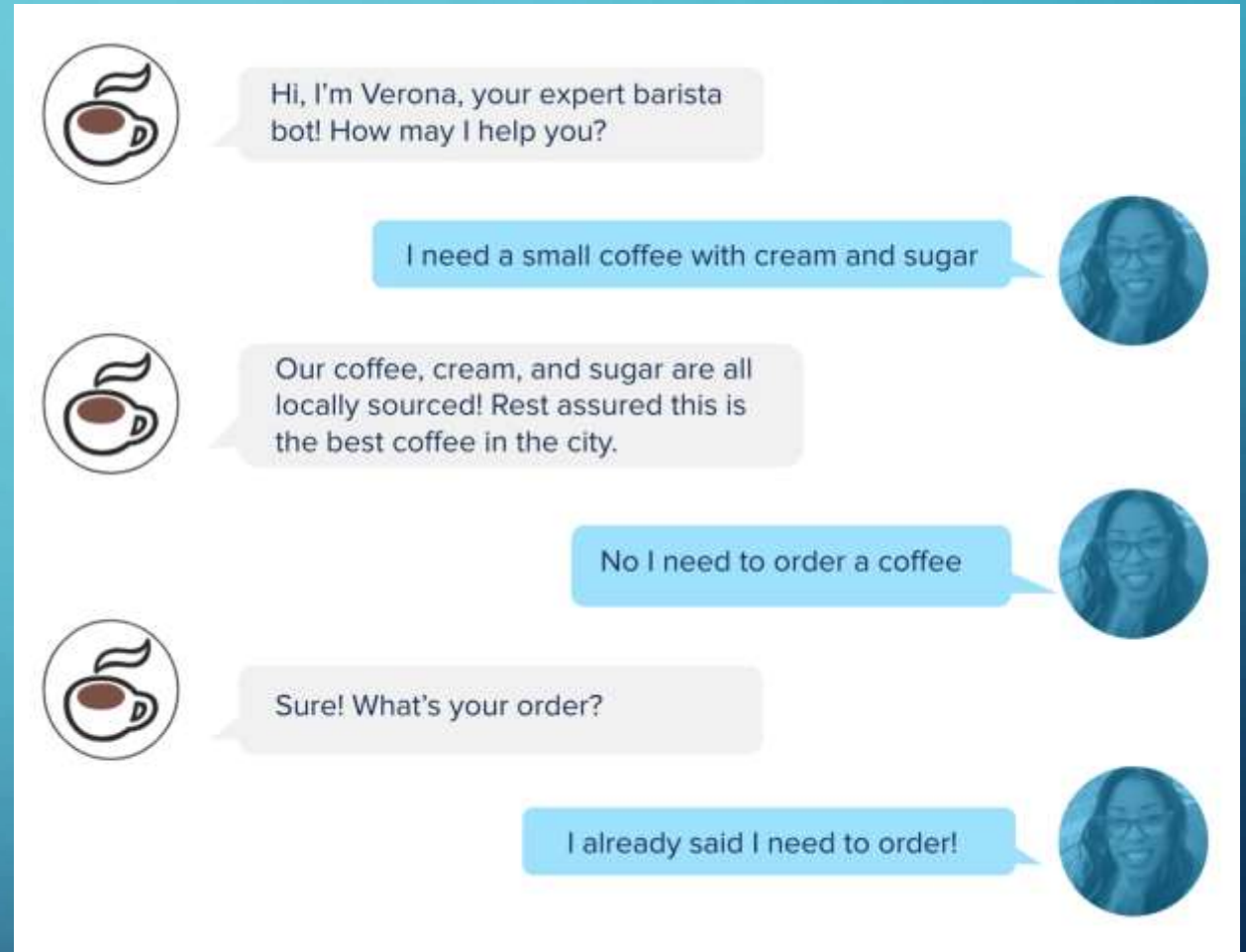
C:\Users\limited\Desktop>es -size -dm -sizecolor 13 -dmcolor 11 -sort dm -n 10
 78,440 26/10/2016 20:02 C:\Users\limited\Desktop\es.exe
2,023,424 26/10/2016 13:57 C:\Users\limited\Desktop\Everything.exe
 32,768 21/10/2016 16:37 C:\Boot\BCD
1,048,576 21/10/2016 16:37 C:\Users\limited\NTUSER.DAT
 53,767 21/10/2016 16:37 C:\Users\limited\AppData\Local\IconCache.db
268,435,456 21/10/2016 16:08 C:\swapfile.sys
738,197,504 21/10/2016 16:08 C:\pagefile.sys
387,332 19/10/2016 19:31 C:\Users\limited\Desktop\Everything-Debug.txt
 181 19/10/2016 18:47 C:\Users\limited\Desktop\Everything.ini
7,328,537 14/10/2016 23:25 C:\dev\Everything.db

C:\Users\limited\Desktop>_
```

Command Line Interface (CLI)

CONVERSATIONAL USER INTERFACE

A conversational user interface (CUI) is a digital interface that enables users to interact with software following the principles of human-to-human conversation



USER INTERFACE DESIGN

User interfaces enable users to effectively control the computer. A successful user interface should be efficient and user-friendly

ELEMENTS OF A USER INTERFACE

- INPUT CONTROLS
- NAVIGATION
- INFORMATIONAL COMPONENTS
- CONTAINERS

GOOD USER DESIGN

- Increase efficiency
- Improves productivity
- Reduces errors
- Reduces training
- Improves acceptance



GOOD USER DESIGN

1. **USEFUL**

- accomplish what is required
- functional, does things

2. **USABLE**

- easy to do things, does the right things, enjoyable, user experience

3. **USED**

- make people want to use it
- attractive, available, acceptable to organisation



IMPORTANCE OF GOOD DESIGN

- **Good design makes a strong first impression.**
- **Good design helps your business stand out.**
- **Good design builds customer relationships.**
- **Good design keeps your website easy to use.**
- **Good design promotes brand consistency.**
- **Good design is the key to social media success.**
- **Good design makes you money**

A Brief History of Screen Design

- In early 1970s, Screen design at this time period had little to guide it, being driven by hardware and telephone line transmission issues.
- A 1970s screen often resembled the one pictured in Figure 1.1.
- Effectively using this kind of screen required a great deal of practice and patience.
- Most early screens were monochromatic, typically presenting green text on black backgrounds.

TDX95210		THE CAR RENTAL COMPANY			10/11/76 10:25	
NAME		TEL		RO		
_____		_____		_____		
PUD	RD	C	RT	MPD		
_____	_____	_____	_____	_____		
ENTRY ERROR XX465628996Q.997						
Command==>						

Figure 1.1 A 1970s screen.

A Brief History of Screen Design

- At the turn of the decade guidelines for text-based screen design were finally made widely available (Galitz, 1980, 1981) and many screens began to take on a much less cluttered look through concepts such as grouping and alignment of elements, as illustrated in Figure 1.2.

THE CAR RENTAL COMPANY

RENTER >> Name: _____
 Telephone: _ _ _ _

LOCATION >> Office: _____
 Pick-up Date: _ _ _
 Return Date: _ _ _

AUTOMOBILE >> Class: _____ (PR, ST, FU, MD, CO, SC)
 Rate: _____
 Miles Per Day: _____

The maximum allowed miles per day is 150.

Enter F1=Help F3=Exit F12=Cancel

Figure 1.2 A 1980s screen.

A Brief History of Screen Design

- The advent of graphics yielded another milestone in the evolution of screen design, as illustrated in Figure 1.3. While some basic "design principles did not change, groupings and alignment, for example, borders were made available to visually enhance groupings, and buttons and menus for implementing commands replaced function keys.
- The entry field was supplemented by a multitude of other kinds of controls, including list boxes, drop-down combination boxes, spin boxes, and so forth.

THE CAR RENTAL COMPANY

RENTER

Name:

Telephone:

LOCATION

Office:

Pick-up Date:

Return Date:

AUTOMOBILE

Class:

Rate:

Miles Per Day:

OK Apply Cancel Help

Figure 1.3 A 1990s and beyond screen.

DIRECT MANIPULATION

Direct manipulation (DM) is an interaction style in which users act on displayed objects of interest using physical, incremental, reversible actions whose effects are immediately visible on the screen.

CHARACTERISTICS

- Physical actions instead of complex syntax
- Continuous feedback
- Rapid learning
- Representations of the objects

GUI

Digital interface in which a user interacts with graphical components such as icons, buttons and menus

CHARACTERISTICS

- VISUALIZATION
- OBJECT ORIENTATION
- AESTHETIC
- ATTRACTIVE
- SCREEN BASED
- CONTROLS



ADVANTAGES

- Easiness for non-technical people
- Looks nicer than text interface
- User-friendly
- Eye Pleasing



DISADVANTAGES

- Slower than command line tools
 - Difficult to develop
 - Memory Consumption
 - Extra Attention required
- 

WEB USER INTERFACE

A Web user interface or Web app allows the user to interact with content or software running on a remote server through a Web browser. The content or web pages is downloaded from the Web server and the user can interact with this content in a Web browser, which acts as a client.

The screenshot displays the Exinda Service Delivery Point (SDP) web user interface. The interface includes a top navigation bar with links for Setup, Tools, Help, and Logout. A left sidebar contains a 'Select Domain' dropdown menu, 'Recently viewed Items', and a 'Dynamic Menu'. The main content area shows a welcome message, instructions for activating the SDP agent, and a list of recent items. A right sidebar displays an 'Appliance Status Summary' table and 'Company Statistics (2008)'.

Annotations with red arrows point to various UI elements:

- Click to view Dashboard
- Click to view all Appliances
- Click to view Reports
- Click to view the Config Log
- Click to access the Setup menu
- Click to access the Tools menu
- Click to access this SDP user guide at any time
- Click to sign out of your SDP session
- SDP version number and release date
- Appliance status summary table
- Total reduction statistics table
- Click to fill in a feedback form
- Select a Specific domain from the drop down menu
- Recently viewed Items

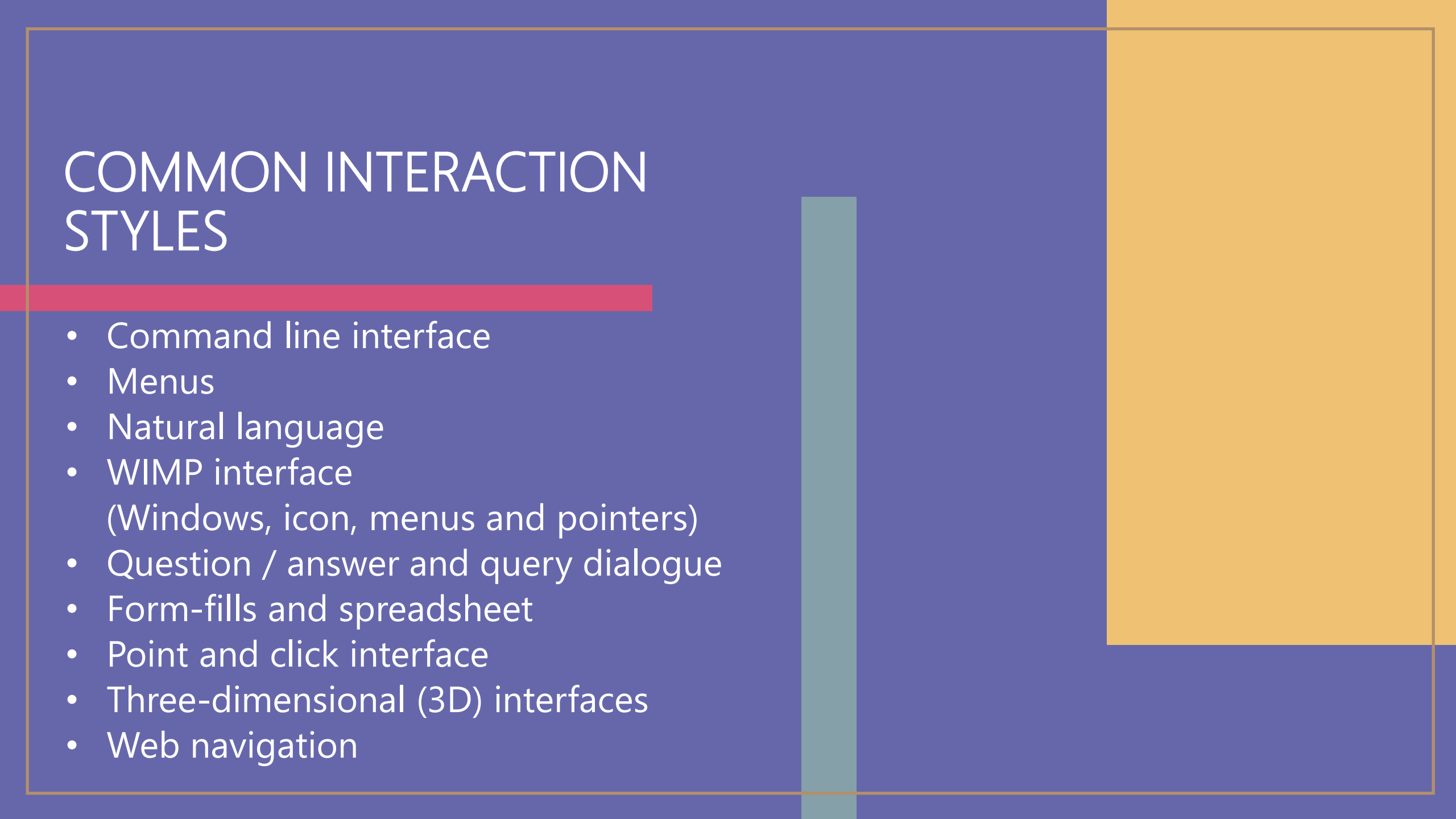
Appliance Status Summary

Status	Count
Online	6
Offline	19
Unlicensed	1

Company Statistics (2008)

Category	Value
Pre-Reduction	3.79 TB
Post-Reduction	102.55 GB
Total Reduction	3.69 TB
Ratio (97.23%)	

COMMON INTERACTION STYLES



- Command line interface
- Menus
- Natural language
- WIMP interface
(Windows, icon, menus and pointers)
- Question / answer and query dialogue
- Form-fills and spreadsheet
- Point and click interface
- Three-dimensional (3D) interfaces
- Web navigation

MENUS

- Menu interface: commands organized into logical groups (more intuitive than command)
- A submenu can be used to present further related list of sub-functions or options
- Menu structure limits range of options (less expressive than command entry)
- Restricted form if WIMP



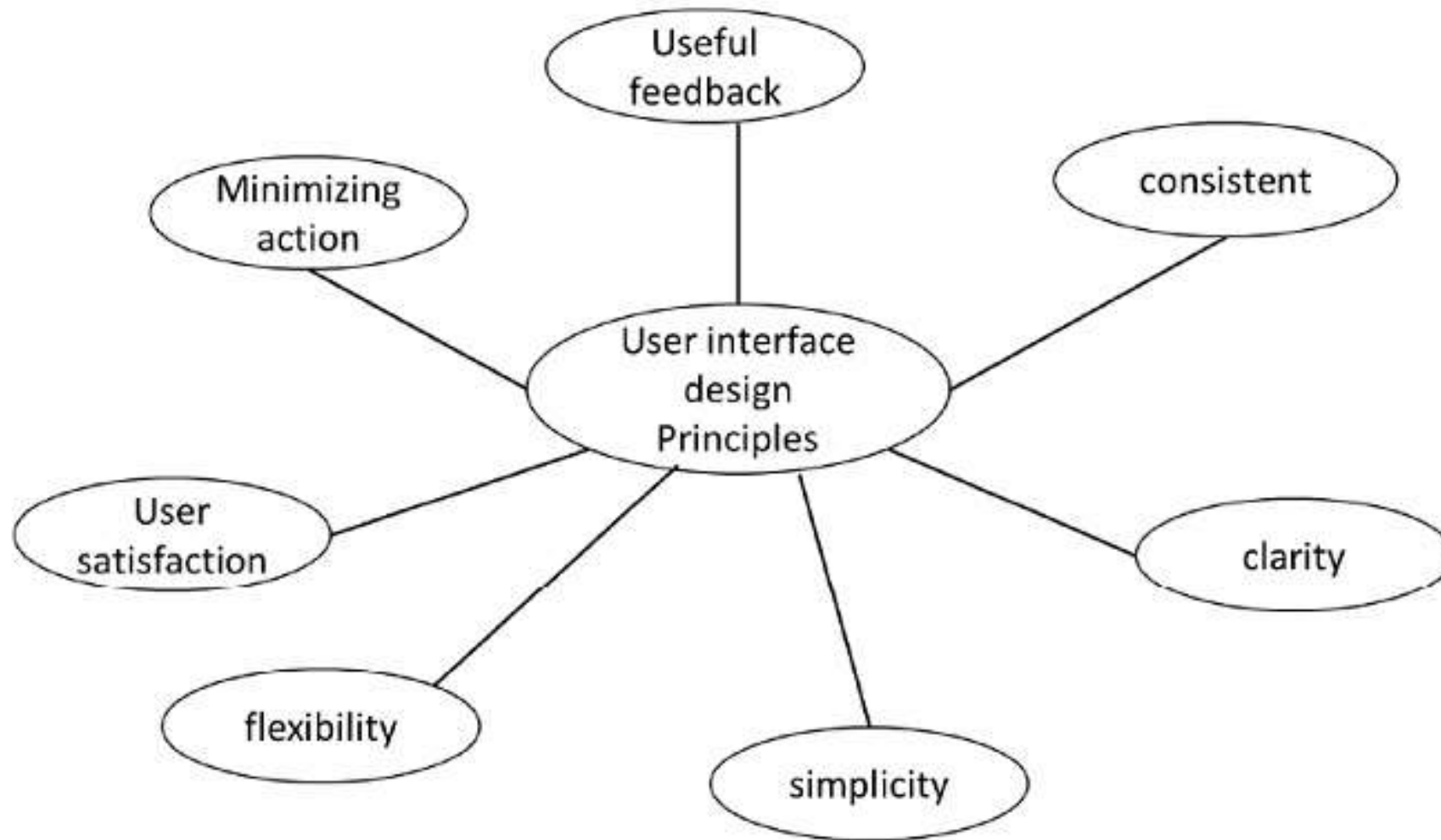
FORM INTERFACE

- Form Interface: presents specific questions to which user must respond in order to perform some task.
- Intuitive, since users are led step by step through interaction
- Not expensive, since form allows to only a few specialized commands

WIMP

- WIMP stands for Windows, Icons, Menus, Pointers
- WIMP interfaces are familiar as they are the basis of most desktop computer operating systems

PRINCIPLES OF USER INTERFACE



- 
- The background of the slide is a dark blue gradient. It is decorated with stylized, light blue circuit-like lines. These lines are composed of straight segments and small circles, resembling a network or a set of data paths. They are located primarily along the left and right edges of the slide, with some lines extending slightly into the central area.
- **Minimize actions** – Minimize action means steps per screen. The tasks and actions are streamlined so that they can be done in as few steps as possible. The interface should be designed keeping in mind to maintain the steps as few as possible for performing any tasks.
 - **Simplicity** – The user interface should not be complex. It should always be designed simple and elegant.
 - **Consistent** – The user interface should be consistent. The design should be consistent. Increasing consistency increases the familiarity, and hence increases the usability.

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- **Clarity** – Content should provide the user with clarity. There should not be anything which confuses the user, as it becomes an obstacle for the user in interacting with the product.
 - **Providing useful feedback** – The user should be provided with feedback for every action. This keeps the user informed and helps them to know whether some action was successful or not.
- 
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JOB ROLES

- FRONT-END DESIGNER
- USER INTERFACE (UI) DESIGNER
- USER EXPERIENCE (UX) DESIGNER
- GRAPHIC DESIGNER
- ANIMATION DESIGNER
- BRAND AND PRODUCT DESIGNER

Introduction of the Graphical User Interface

Table 1.2 Chronological History of Graphical User Interfaces

1973	Pioneered at the Xerox Palo Alto Research Center. —First to pull together all the elements of the modern GUI.
1981	First commercial marketing as the Xerox STAR. —Widely introduced pointing, selection, and mouse.
1983	Apple introduces the Lisa. — Features pull-down menus and menu bars.
1984	Apple introduces the Macintosh. — Macintosh is the first successful mass-marketed system.
1985	Microsoft Windows 1.0 released. Commodore introduces the Amiga 1000.
1987	X Window System becomes widely available. IBM's System Application Architecture released. — Including Common User Access (CUA). IBM's Presentation Manager released. — Intended as graphics operating system replacement for DOS. Apple introduces the Macintosh II. — The first color Macintosh.
1988	NeXT's NeXTStep released. — First to simulate three-dimensional screen.
1989	UNIX-based GUIs released. — Open Look by AT&T and Sun Microsystems. — Innovative appearance to avoid legal challenges. — Motif, for the Open Software Foundation by DEC and Hewlett-Packard. — Appearance and behavior based on Presentation Manager. Microsoft Windows 3.0 released.
1992	OS/2 Workplace Shell released. Microsoft Windows 3.1 released.
1993	Microsoft Windows NT released.
1995	Microsoft Windows 95 released.
1996	IBM releases OS/2 Warp 4. Microsoft introduces NT 4.0.
1997	Apple releases the Mac OS 8.
1998	Microsoft introduces Windows 98.
1999	Apple releases Mac OS X Server. — A UNIX-based OS.
2000	Microsoft Windows 2000 released. Microsoft Windows ME released
2001	Microsoft Windows XP released

The Blossoming of the World Wide Web

Table 1.3 Chronological History of the Internet

1945	Hypertext concept presented by Vannevar Bush.
1960	J. C. R. Licklider of MIT proposes a global network of computers.
1962	Design and development begins on network called ARPANET
1969	ARPANET is brought online. — Connects computers at four major universities. — Additional universities and research institutions soon added to the network.
1973	ARPANET goes international.
1974	Bolt, Beranek and Newman releases Telenet. — The first commercial version of ARPANET.
1976	University of Vermont's PROMIS released. — The first hypertext system released to the user community.
1982	The term Internet is coined.
1983	TCP/IP architecture now universally adopted.
1988	Apple's HyperCard released. —Presents the hypertext idea to a wider audience. —The first Internet worm unleashed.
1989	Tim Berners-Lee and others at the European Laboratory for Particle Physics (CERN) propose a new protocol for distributing information. — Based upon hypertext.
1990	HTML created. — In conjunction with Berners-Lees protocol. ARPANET is decommissioned
1991	HTML code released on the Internet by Tim Berners-Lee. Berners-Lee's work is credited with hatching the World Wide Web. Gopher developed at the University of Minnesota. — First really friendly interface.

The Blossoming of the World Wide Web

Table 1.3 (Continued)

1992	Delphi released. – First to provide commercial online Internet access to subscribers. Mosaic created by the National Center for Supercomputing Applications (NCSA) at the University of Illinois. – The first popular graphic-based hypertext browser.
1994	Netscape Navigator Version 1.0 released. World Wide Web Consortium founded. – To promote and develop Web standards.
1995	Microsoft Internet Explorer Versions 1.0 and 2.0 released. AOL, CompuServe, Prodigy, Yahoo, and Lycos come online. National Science Foundation ends Internet support. HTML 2.0 approved as proposed Web standard. Netscape Navigator Versions 2.0 and 3.0 released. Microsoft Internet Explorer Version 3.0 released. Opera Version 2.1 released. – Browser for computers with small resources. – Written from scratch (not based upon Mosaic). – Version 2.1 the first widely available. HTML 3.2 draft released. NCSA halts development of Mosaic. Netscape Navigator Version 4.0 released. Microsoft Internet Explorer Version 4.0 released. Opera Version 3.0 released. HTML 4.0 certified as proposed standard. Microsoft Internet Explorer Version 5.0 released. XHTML 1.0 first working draft released.

HCI GOALS

- The goal of HCI is to improve the interaction between users and computers by making computers more user-friendly and receptive to the user's needs.
- Improving the usability is the key factor
- To design systems that minimize the barrier between the human's cognitive model of what they want to accomplish and the computer's understanding of the user's task



